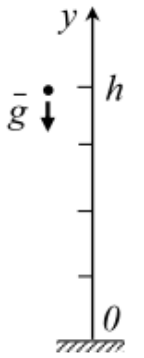
Fizika yechimlari

1. Biror balandlikdan boshlang'ich tezliksiz erkin tushayotgan jism yo'lning birinchi ¼ qismini 1 s da bosib o'tdi. Jism yo'lning qolgan qismini qancha vaqtda bosib o'tadi?

Berilgan: $t_1 = 1\ s,\ g = 9{,}8\ m/s^2$;
Topish kerak: $\Delta t - ?$
Masalaning yechilishi:



1-rasm

$$y = y_0 - \frac{gt^2}{2} = \frac{h}{4} - \frac{gt_1^2}{2}$$

$$\frac{h}{4} = \frac{gt_1^2}{2}\ h = 2gt_1^2$$

$$y = y_0 - \frac{gt^2}{2} = h - \frac{gt_2^2}{2}$$

$$h = \frac{gt_2^2}{2} \quad t_2 = \sqrt{\frac{2h}{g}} = 2t_1\ \Delta t = t_2 - t_1 = t_1$$

$$\boxed{\Delta t = t_1}$$

2. Ikki jism bir xil boshlang'ich tezlik bilan, 2 s oralatib yuqoriga tik otildi. Agar ikkinchi jism otilgan paytdan 1,5 s o'tgach jismlarning tezliklari modul jihatdan tenglashgan bo'lsa, ularning boshlang'ich tezligini aniqlang.

Berilgan: $\Delta t = 2\ s,\ t = 1{,}5\ s,\ g = 9{,}8\ m/s^2$;
Topish kerak: $\upsilon_0 - ?$
Masalaning yechilishi:

1–usul. *Oy* o'qini vertikal tarzda yuqoriga yo'naltirib, koordinata boshini Yer bilan bog'laymiz (2 –rasm). Harakat vaqtini ikkinchi jism otilgan paytdan hisoblaymiz. Jismlarning oniy tezligini topish uchun $\bar{\upsilon} = \bar{\upsilon}_0 + \bar{g}\,t$ ifodani tatbiq etamiz:

$$\upsilon_1 = \upsilon_0 - g(t + \Delta t) \qquad (1)$$
$$\upsilon_2 = \upsilon_0 - gt \qquad (2)$$

Jismlar uchrashgan nuqtada ularning tezliklarini moduli teng bo'ladi, ya'ni: $|\upsilon_1| = |\upsilon_2|$. (1) va (2) ifodalardan:

$$\upsilon_0 - g(t + \Delta t) = -(\upsilon_0 - gt)$$

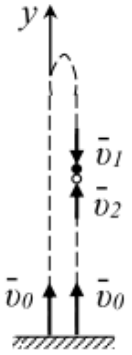
Bundan

$$\boxed{\upsilon_0 = \frac{(2t + \Delta t)\, g}{2}} \qquad (3)$$

2–usul. *Oy* o'qini vertikal tarzda yuqoriga yo'naltirib, koordinata boshi sifatida yer sirtini olamiz. Harakat vaqtini ikkinchi jism otilgan paytdan hisoblaymiz. U holda jismlarning koordinatalari quyidagi qonuniyat bo'yicha o'zgarib boradi:

$$y_1 = \upsilon_0(\Delta t + t) - \frac{g(\Delta t + t)^2}{2} \qquad (1)$$

$$y_2 = \upsilon_0 t - \frac{gt^2}{2} \qquad (2)$$



2-rasm

Agar jismlarning tezliklari modul jihatdan tenglashgan bo'lsa, shu onda ularning koordinatalari bir xil bo'ladi, ya'ni: $y_1 = y_2$. (1) va (2) ifodalarning o'ng tomonidagi hadlarni tenglaymiz:

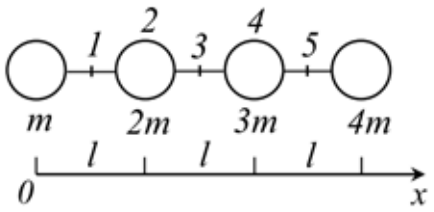
$$\upsilon_0 t - \frac{gt^2}{2} = \upsilon_0(\Delta t + t) - \frac{g(\Delta t + t)^2}{2}$$

Bundan

$$\boxed{\upsilon_0 = \frac{(2t + \Delta t)\, g}{2}} \qquad (3)$$

Hisoblash natijasi: $\upsilon_0 = 25\ m/s$.

3. Qaysi nuqtaga tayanch qo'yilsa, jismlar sistemasi muvozanatda bo'ladi?



(3-rasm)

Masalaning yechilishi:

Ox o'qini gorizont bo'ylab yo'naltiramiz. Koordinata boshi sifatida *m* massali sharning geometrik markazini olamiz. Jismlar orasidagi masofa *l* ga teng bo'lsin (3 –rasm). Jismlar sistemasining massa markazini aniqlash ifodasidan foydalanamiz. U holda

$$x_{mm} = \frac{\sum m_i x_i}{\sum m_i} = \frac{2ml + 3m2l + 4m3l}{m + 2m + 3m + 4m} = \frac{20ml}{10m} = 2l$$

Massa markazining *Oy, Oz* o'qlari bo'yicha koordinatasi uchun quyidagi ifoda o'rinli:

$$y_{mm} = 0,\ z_{mm} = 0$$

Demak, sistemaning massa markazi ularni geometrik markazlarini tutashtiruvchi to'g'ri chiziq bo'ylab, *m* massali jism markazidan *2l* masofada joylashgan *4* nuqtada bo'ladi.

@olimpdep



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